# WORKSHEET — UNIT II

## Quantum Gates and Circuits

**Course:** Fundamentals of Quantum Computing

**Unit:** II (5 Hrs / 300 min)

**Topics:** From States to Operations; Reversible Computing and the Unitary Requirement; Pauli Gates; Hadamard, Phase and T Gates; Multi-Qubit Gates; The Quantum Circuit Model; Universal Gate Sets; Worked Circuit Designs and Simulation

## 🎯 Section A — Learning Objectives

**1.** Explain why classical AND/OR are irreversible, state Landauer's principle, and show why every quantum gate must be a unitary matrix.

**2.** Write the matrices and actions of the Pauli gates X, Y, Z and the rotation gates Rx(θ), Ry(θ), Rz(θ), and use the Pauli algebra.

**3.** Use the Hadamard, S, T and phase gates, and apply identities such as HXH = Z, $S^2 = Z$ and $T^2 = S$.

**4.** Describe CNOT, controlled-Z, SWAP, Toffoli, Fredkin and controlled-U gates, and explain how H + CNOT creates a Bell state.

**5.** Read and compose quantum circuits (series and parallel), measure depth, width and gate count, and compute a 2-qubit output by matrix multiplication.

**6.** Explain universal gate sets, the Solovay–Kitaev theorem and transpilation, and design Bell, half-adder, teleportation and superdense-coding circuits.

## 🔑 Section B — Key Concepts

| No. | Concept | Short Explanation |
|---|---|---|
| 1 | **Reversibility and Unitarity** | Quantum evolution is reversible, so every gate is a unitary matrix with U†U = I; this keeps the total probability equal to 1. |
| 2 | **Reversible Classical Gates** | NOT, CNOT, Toffoli and Fredkin lose no information, unlike AND and OR; Landauer's principle ties erased information to heat. |
| 3 | **Pauli Gates** | X flips $\|0\rangle$ and $\|1\rangle$, Z flips the sign of $\|1\rangle$ (a phase flip), and Y does both; each is a π rotation on the Bloch sphere. |
| 4 | **Rotation Gates** | Rx(θ), Ry(θ) and Rz(θ) rotate the Bloch vector by angle θ about the x, y or z axis; they generalize the Pauli gates. |
| 5 | **Hadamard Gate** | H turns $\|0\rangle$ into $\|+\rangle$ and $\|1\rangle$ into $\|-\rangle$; it switches between the Z-basis and the X-basis and creates superposition. |
| 6 | **Phase Gates S, T and P(φ)** | These leave $\|0\rangle$ unchanged and multiply $\|1\rangle$ by a phase: i for S, e^{iπ/4} for T and e^{iφ} for P(φ). |
| 7 | **Controlled Gates and CNOT** | A control qubit decides whether a gate acts on the target; CNOT flips the target when the control is $\|1\rangle$. |
| 8 | **SWAP, Toffoli and Fredkin** | SWAP exchanges two qubits (three CNOTs); Toffoli flips a target when two controls are 1; Fredkin swaps two targets when one control is 1. |
| 9 | **Quantum Circuit Model** | Wires carry qubits left to right in time; gates in series multiply as matrices and gates in parallel combine by the tensor product. |
| 10 | **Universal Gate Sets** | Small sets such as {H, T, CNOT} can approximate any unitary; the Solovay–Kitaev theorem says the approximation is efficient. |
| 11 | **Worked Circuit Designs** | Bell generator, half-adder, teleportation and superdense coding show how gates combine to perform useful tasks. |

## 📘 Section C — Important Terminology

| Keyword | Meaning |
|---|---|
| **Reversible Computation** | Computation in which the input can always be recovered from the output, so no information is erased. |
| **Landauer's Principle** | Erasing one bit of information dissipates at least kT ln 2 of heat. |
| **Unitary Gate** | A gate whose matrix satisfies U†U = I; it preserves norms and probability. |
| **Pauli Matrices** | The matrices X, Y and Z (with the identity I), which generate all single-qubit operators. |
| **Anticommutation** | Two operators A and B anticommute when AB = −BA; distinct Pauli gates anticommute. |
| **Rotation Gate** | A gate Rn(θ) that rotates the Bloch vector by θ about the axis n. |
| **Hadamard Gate (H)** | The gate (1/√2)[[1, 1], [1, −1]] that maps \|0⟩ to \|+⟩ and \|1⟩ to \|−⟩. |
| **S Gate** | The phase gate diag(1, i), a quarter turn about the z-axis; $S^2 = Z$. |
| **T Gate (π/8)** | The phase gate diag(1, e^{iπ/4}); $T^2 = S$. |
| **Control and Target** | In a controlled gate, the control qubit decides whether the operation is applied to the target qubit. |
| **CNOT** | The two-qubit gate that flips the target when the control is \|1⟩. |
| **Controlled-Z (CZ)** | A two-qubit gate that applies Z to the target when the control is \|1⟩; it flips the sign of \|11⟩. |
| **SWAP Gate** | A two-qubit gate that exchanges the states of two qubits. |
| **Toffoli (CCNOT)** | A three-qubit gate that flips the target only when both controls are \|1⟩. |
| **Fredkin (CSWAP)** | A three-qubit gate that swaps two target qubits when the control is \|1⟩. |
| **Circuit Depth** | The number of sequential layers of gates in a circuit. |
| **Circuit Width** | The number of qubits (wires) used by a circuit. |
| **Gate Count** | The total number of gates in a circuit. |
| **Deferred Measurement** | Measurements can be moved to the end of a circuit, replacing classical control with quantum control, without changing the results. |
| **Universal Gate Set** | A finite set of gates from which any unitary can be approximated to any accuracy. |
| **Solovay–Kitaev Theorem** | Any single-qubit unitary can be approximated to accuracy ε using only polylogarithmically many gates in 1/ε from a universal set. |
| **Clifford Group** | The group generated by H, S and CNOT; Clifford circuits are not universal on their own and become universal when T is added. |
| **Transpilation** | Rewriting a circuit into the native gates and connectivity of a specific quantum device, often while reducing depth. |
| **Native Gates** | The small set of gates a hardware device actually implements; all other gates are built from them. |

## ✅ Section D — Multiple-Choice Questions [Easy–Medium]

**1. Why must every quantum gate be a unitary matrix?**

**A.** To make the gate run faster
**B.** To keep every matrix diagonal
**C.** To preserve total probability and keep the operation reversible
**D.** To avoid using complex numbers

**2. Which classical gate is irreversible?**

**A.** AND
**B.** NOT
**C.** CNOT
**D.** Toffoli

**3. According to Landauer's principle, what is the minimum cost of erasing one bit of information?**

**A.** No energy at all
**B.** Exactly one joule
**C.** Energy that depends only on the number of gates
**D.** At least kT ln 2 of heat

**4. What is the result of applying the Pauli-Z gate to $|+\rangle$?**

**A.** $|+\rangle$
**B.** $|-\rangle$
**C.** $|0\rangle$
**D.** $|1\rangle$

**5. Which identity is correct?**

**A.** HXH = Y
**B.** HZH = Z
**C.** HXH = Z
**D.** $S^2 = X$

**6. The T gate squared, $T^2$, equals which gate?**

**A.** S
**B.** Z
**C.** X
**D.** H

**7. A CNOT with the first qubit as control acts on $|10\rangle$. What is the result?**

**A.** $|10\rangle$
**B.** $|00\rangle$
**C.** $|01\rangle$
**D.** $|11\rangle$

**8. How many CNOT gates are needed to build a SWAP gate?**

**A.** 2
**B.** 3
**C.** 4
**D.** 1

**9. What does the depth of a quantum circuit measure?**

**A.** The number of sequential layers of gates
**B.** The number of qubits used
**C.** The total number of gates
**D.** The number of measurements

**10. Which gate set is universal for quantum computation?**

**A.** {X, Y, Z}
**B.** {H, S, CNOT}
**C.** {H, T, CNOT}
**D.** {Z, S, T}

## ✏️ Section E — Fill in the Blanks [Easy]

**1.** A gate U is unitary when U†U equals the _____ matrix.

**2.** The Hadamard gate maps $|0\rangle$ to the superposition state _____.

**3.** The Pauli-_____ gate is the phase-flip gate that turns $|+\rangle$ into $|-\rangle$.

**4.** The three-qubit gate that flips its target only when both controls are 1 is the _____ gate.

**5.** The gate set {H, T, CNOT} is called a _____ gate set because it can approximate any unitary.

## ☑️ Section F — True or False [Easy]

**1.** The classical AND gate is reversible. (True / False)

**2.** For the Pauli gates, $X^2 = Y^2 = Z^2 = I$. (True / False)

**3.** Applying the Hadamard gate twice returns the original state. (True / False)

**4.** A CNOT gate acting on $|00\rangle$ alone can create an entangled state. (True / False)

**5.** Measurements can be deferred to the end of a circuit without changing the outcome statistics. (True / False)

## 🔗 Section G — Match the Following [Medium]

| Column A | Column B (shuffled) |
|---|---|
| 1. Pauli-X | a. Exchanges two qubits and equals three CNOT gates |
| 2. Hadamard gate | b. Switches between the Z-basis and the X-basis and creates superposition |
| 3. T gate | c. Swaps two target qubits when the control is 1 |
| 4. SWAP gate | d. Quantum NOT: flips $\|0\rangle$ and $\|1\rangle$ (a π rotation about the x-axis) |
| 5. Toffoli gate | e. Phase gate diag(1, e^{iπ/4}), costly in fault-tolerant machines |
| 6. Fredkin gate | f. Any single-qubit unitary can be approximated efficiently from a universal set |
| 7. Solovay–Kitaev theorem | g. Flips the target when both controls are 1 |

## 📝 Section H — Short-Answer Questions [Medium]

**1. Why must quantum gates be unitary, and what does this imply about reversibility?**

**2. Write the matrices of X and Z, and state their effect on $|0\rangle$ and $|+\rangle$.**

**3. Show that H switches between the Z-basis and the X-basis, and verify that HXH = Z.**

**4. Give the truth table of CNOT and explain how H followed by CNOT produces a Bell state from $|00\rangle$.**

**5. What is a universal gate set, and why are T gates expensive and transpilation necessary on real hardware?**

## 📄 Section I — Descriptive Questions [Medium–Advanced]

**1. Explain reversible computing: why AND and OR are irreversible, Landauer's principle, the reversible gates NOT, CNOT, Toffoli and Fredkin, and how to verify that a given matrix is unitary.**

**2. Describe the single-qubit gates: the Pauli family and its algebra, the rotation gates Rx, Ry and Rz, the Hadamard, S, T and phase gates, the identities among them, and how an arbitrary single-qubit unitary is built from rotations.**

**3. Explain multi-qubit gates and the circuit model: controlled gates, CNOT, controlled-Z, SWAP, Toffoli, Fredkin and controlled-U, composition in series and in parallel, depth, width and gate count, and the deferred-measurement principle.**

**4. Discuss universal gate sets and worked designs: what universality means, {H, T, CNOT}, the Solovay–Kitaev theorem, Clifford+T, transpilation, and the Bell-state, half-adder, teleportation and superdense-coding circuits.**

## 💻 Section J — Practical / Application-Based Questions [Advanced]

### Practical 1

Problem: Write a Python function that checks whether a 2×2 matrix is a valid quantum gate.

**Task:**

**1.** Define is_unitary(M) that computes the conjugate transpose of M and multiplies it by M.

**2.** Return True when the product equals the 2×2 identity matrix within 1e-9.

**3.** Test it with X = [[0, 1], [1, 0]], the Hadamard matrix and N = [[1, 1], [0, 1]], and print each result.

**Expected Output:** *Three lines: True, True, False.*

### Practical 2

Problem: Apply a sequence of single-qubit gates to a state vector.

**Task:**

**1.** Define apply(G, v) that multiplies a 2×2 gate by a 2-element state vector.

**2.** Start from $|0\rangle$ = [1, 0] and apply H, then Z, then H, printing the state after each gate rounded to 2 decimal places.

**3.** Compare the final state with $X|0\rangle$ and note which identity this shows.

**Expected Output:** *Three lines: (0.71, 0.71), (0.71, -0.71), (0.0, 1.0); the result equals $X|0\rangle$ because HZH = X.*

### Practical 3

Problem: Compute the output state of a 2-qubit circuit by matrix multiplication.

**Task:**

**1.** Write kron(A, B) that returns the 4×4 Kronecker product of two 2×2 matrices, and store the CNOT matrix with the first qubit as control.

**2.** Start from $|00\rangle$ = [1, 0, 0, 0] and apply H ⊗ I, then CNOT.

**3.** Print the four amplitudes rounded to 2 decimal places and say which Bell state results.

**Expected Output:** *One line: [0.71, 0.0, 0.0, 0.71], the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$.*

## 🗺️ Section K — Concept-Mapping Activity

**QUANTUM GATES AND CIRCUITS**

*Complete the map below: fill in the missing sub-topic name or example for each branch.*

| Branch 1 | Branch 2 | Branch 3 | Branch 4 |
|---|---|---|---|
| **Reversible Computing** | **Single-Qubit Gates** | **Multi-Qubit Gates and Circuits** | **Universality and Designs** |
| Example: _____ | Example: _____ | Example: _____ | Example: _____ |

*(Answer key — example values: Landauer's principle, Toffoli gate, unitarity; Pauli-X, Hadamard, T gate; CNOT, circuit depth, deferred measurement; Clifford+T, transpilation, teleportation)*

## 🔬 Section L — Case Study / Real-World Activity

### Scenario

A student team must (a) generate a Bell pair, (b) build a quantum half-adder that adds two one-bit inputs a and b into a sum bit and a carry bit, and (c) run both circuits on a real quantum device. The device supports only a small native set of single-qubit rotations and one two-qubit gate, and its qubits lose their state quickly, so long circuits give noisy results.

### Questions

**1.** Give the gate sequence of the Bell-pair circuit on $|00\rangle$, its output state, and the circuit's width, depth and gate count.

**2.** Design the half-adder with CNOT and Toffoli gates, and give the sum and carry for a = 1 and b = 1.

**3.** The device does not support H or Toffoli directly. What process rewrites the circuits, and what two quantities does it try to reduce?

**4.** Why does circuit depth matter more on real hardware than on a simulator?

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### Expected Learning Outcome

*Students design small circuits from standard gates and reason about the effect of native gate sets, transpilation and depth on real hardware.*

## 📊 Section M — Self-Assessment

☐ **Yes** ☐ **Partly** ☐ **No** I can explain the major concepts of this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can define the important terminology from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can solve problems/exercises from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can apply the concepts of this unit to a new situation.

## 💭 Section N — Student Reflection

**What did I learn in this unit?**

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**What was difficult for me?**

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**What do I need to practice more?**

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**One important concept I learned:**

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## 🗝️ Section O — Instructor Answer Key

*(For instructor use — covers all objective and subjective items in this worksheet.)*

### MCQ Answers

- 1. C — Unitarity preserves the norm (total probability 1) and guarantees an inverse U†, so the operation is reversible.
- 2. A — AND maps several inputs to one output, so the inputs cannot be recovered; NOT, CNOT and Toffoli are reversible.
- 3. D — Landauer's principle: erasing one bit dissipates at least kT ln 2 of heat.
- 4. B — Z flips the sign of the $|1\rangle$ component, so $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ becomes $|-\rangle$.

- 5. C — HXH = Z and HZH = X; $S^2 = Z$, so D is wrong.
- 6. A — $T^2 = S$, because the phases e^{iπ/4} multiply to e^{iπ/2} = i.
- 7. D — The control is 1, so the target flips: $|10\rangle \rightarrow |11\rangle$.
- 8. B — SWAP = CNOT(0→1) · CNOT(1→0) · CNOT(0→1).
- 9. A — Depth counts sequential layers; width counts qubits and gate count counts all gates.
- 10. C — {H, T, CNOT} is universal; {H, S, CNOT} is only the Clifford set and {X, Y, Z} and {Z, S, T} cannot create superposition or entanglement.

## Fill in the Blanks — Answers

- 1. identity (I)
- 2. $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$
- 3. Z
- 4. Toffoli (CCNOT)
- 5. universal

## True/False — Answers

- 1. False
- 2. True
- 3. True
- 4. False
- 5. True

## Match the Following — Answer Key

1→d 2→b 3→e 4→a 5→g 6→c 7→f

## Short Answer — Key Points

- 1. Quantum evolution must keep total probability equal to 1, which means U†U = I. A unitary always has an inverse U†, so every gate can be undone; the operation is reversible.
- 2. X = [[0, 1], [1, 0]] and Z = [[1, 0], [0, −1]]. $X|0\rangle = |1\rangle$ and $Z|0\rangle = |0\rangle$; $X|+\rangle = |+\rangle$ and $Z|+\rangle = |-\rangle$.
- 3. $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$, so H swaps the bases. Since H maps X-eigenstates to Z-eigenstates, HXH = Z; check by matrix multiplication.
- 4. CNOT: 00→00, 01→01, 10→11, 11→10. H on the control gives $(|0\rangle + |1\rangle)|0\rangle/\sqrt{2}$, and CNOT turns this into $(|00\rangle + |11\rangle)/\sqrt{2}$, which cannot be written as a product state.
- 5. A universal set can approximate any unitary to any accuracy, e.g. {H, T, CNOT}. T gates are costly because they need expensive magic-state distillation in fault-tolerant machines. Hardware has its own native gates and connectivity, so transpilation rewrites the circuit to match them and cut depth.

## Descriptive Questions — Key Points

- 1. AND and OR lose information (many inputs, one output); Landauer: erasing a bit costs at least kT ln 2. Reversible gates: NOT, CNOT, Toffoli, Fredkin. Unitarity: U†U = I keeps probability. Verification: compute U†U and check it gives the identity; e.g. H†H = I but [[1, 1], [0, 1]] fails.
- 2. Pauli X, Y and Z are π rotations; XY = iZ, YX = −iZ so they anticommute, and $X^2 = Y^2 = Z^2 = I$. Rx, Ry and Rz rotate by θ about each axis. H creates superposition and swaps bases; S = diag(1, i), T = diag(1, e^{iπ/4}), P(φ) = diag(1, e^{iφ}). Identities: HXH = Z, HZH = X, $S^2 = Z$, $T^2 = S$. Any single-qubit unitary is a product of rotations, e.g. Rz·Ry·Rz.
- 3. A controlled gate applies an operation to the target when the control is 1: CNOT, CZ, Toffoli, Fredkin, controlled-U. SWAP = three CNOTs. Series gates multiply as matrices (right to left); parallel gates combine by ⊗. Depth = layers, width = qubits, gate count = gates. Deferred measurement moves measurements to the end. Example: H ⊗ I then CNOT on $|00\rangle$ gives a Bell state.
- 4. Universal sets: {H, T, CNOT} and {Rx, Ry, Rz, CNOT}; the Solovay–Kitaev theorem makes approximation efficient; Clifford+T counts T gates as the main cost; transpilation maps to native gates. Designs: Bell generator (H, CNOT); half-adder (two CNOTs for the sum, a Toffoli for the carry); teleportation (Bell pair, Bell measurement, two classical bits, X/Z corrections); superdense coding (shared Bell pair, Alice applies I, X, Z or ZX, Bob decodes with CNOT and H).

## Practical Activities — Expected Outputs

- 1. Sample logic: multiply the conjugate transpose of M by M and test every entry against the identity with abs(P[i][j] − (1 if i == j else 0)) < 1e-9. The tests print True (X), True (H) and False (N).
- 2. Sample logic: def apply(G, v): return [sum(G[i][k]*v[k] for k in range(2)) for i in range(2)], with s = 1/2**0.5 and H = [[s, s], [s, −s]]. The states are (0.71, 0.71), (0.71, −0.71) and (0.0, 1.0), so HZH|0⟩ = |1⟩ = X|0⟩.
- 3. Sample logic: kron(A, B)[i][j] = A[i//2][j//2] * B[i%2][j%2]; CNOT = [[1,0,0,0],[0,1,0,0],[0,0,0,1],[0,0,1,0]]. H ⊗ I on |00⟩ gives [0.71, 0, 0.71, 0] and CNOT gives [0.71, 0.0, 0.0, 0.71].

## Concept Map — Sample Answers

- Reversible Computing: Landauer’s principle, Toffoli gate, Fredkin gate, unitarity. Single-Qubit Gates: Pauli-X, Y and Z, Hadamard, S, T, rotation gates. Multi-Qubit Gates and Circuits: CNOT, CZ, SWAP, circuit depth and width, deferred measurement. Universality and Designs: {H, T, CNOT}, Solovay–Kitaev, Clifford+T, transpilation, teleportation, superdense coding.

## Case Study — Key Points

- 1. H on qubit 0, then CNOT with qubit 0 as control and qubit 1 as target; the output is (|00⟩ + |11⟩)/√2. Width = 2, depth = 2, gate count = 2.
- 2. With inputs a, b and an output qubit s and carry qubit c starting at 0: CNOT(a → s), CNOT(b → s), Toffoli(a, b → c). For a = 1, b = 1: sum = 0 and carry = 1 (1 + 1 = 10 in binary).
- 3. Transpilation rewrites the circuit into native gates and the device’s connectivity; it tries to reduce circuit depth and gate count (especially two-qubit gates and T gates).
- 4. Real qubits decohere and every gate adds error, so a deeper circuit accumulates more noise before it finishes; a simulator has no such noise.